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Generative AI as a catalyst for critical thinking: reframing students from consumers to co-creators

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This study examined students' AI literacy, perceived learning benefits, and ethical awareness following participation in a structured, AI-integrated classroom activity. The study also investigated the relationships among these constructs and examined whether perceived learning benefits mediate the relationship between AI literacy and ethical awareness. In addition, the study explored whether students' responses differed according to gender, prior AI experience, and frequency of AI use. The study involved 104 undergraduate students who participated in a hands-on classroom activity over two academic semesters. Students used generative AI tools, including ChatGPT, Google Gemini, and Microsoft Copilot, to prepare and deliver short lessons to their classmates, followed by discussions addressing responsible and ethical AI use. Data were collected using a structured questionnaire measuring AI literacy, perceived learning benefits, ethical awareness, and related aspects of students' experiences. Overall, students reported positive perceptions of the learning experience. They strongly agreed that AI made learning easier and more enjoyable ($M = 4.29$), saved time on academic tasks ($M = 4.27$), and helped improve the quality of their work ($M = 4.09$). AI literacy was positively correlated with perceived learning benefits ($r = 0.559$, $p < .001$) and ethical awareness ($r = 0.386$, $p < .001$) and no statistically significant gender differences were found across any of the study domains ($p > .05$). The findings suggest that students' AI literacy is associated with greater perceived learning benefits and ethical awareness. Furthermore, the study examines the potential mediating role of perceived learning benefits in the relationship between AI literacy and ethical awareness. These findings highlight the importance of structured, meaningful classroom activities that encourage students to actively engage with, evaluate, and take responsibility for AI-generated content rather than relying on AI outputs passively.

KEYWORDS

active learning, AI literacy, ethical awareness, generative AI, higher education, perceived learning benefits

Introduction

Background and problem statement

Artificial intelligence is no longer something we just read about in tech reports. It is already here—in our homes, on our phones, in the apps we use every day. And in higher education, generative AI has pushed its way to the center of the conversation. It is quietly forcing teachers, researchers, and university leaders to go back and rethink some pretty

basic questions: What does it mean to teach? What does it mean to learn? How is knowledge actually created these days?

Back in 2020, the Horizon Report listed AI as one of six technologies most likely to reshape higher education. But that was before tools like ChatGPT, Google Gemini, and Microsoft Copilot became widely available. Nowadays, the decision feels much more urgent (Brown et al., 2020).

Generative AI? In simple terms, it is a computer system trained on massive amount of data to produce new content (text, images, code, audio, etc.) based on patterns it trained on (Abdelkader et al., 2024; Mittal et al., 2024). Students today use these tools to draft papers, get quick feedback, summarize difficult readings, or look at problems from new angles. And this is where we need to have a stop. On one hand, AI can bring easy, fast, and expert academic support levels to students. This is the positive side. However, on the other hand, there are a real risk: students may become passive receivers of automated machines rather than active, curious, critical thinker learners (Bansal et al., 2024).

Researchers have been documenting both the pros and cons of generative AI in higher education. Some studies show that AI tools can make teaching more efficient, personalize learning, and support student engagement (Bansal et al., 2024; Cordero et al., 2025; Rusmiyanto et al., 2023). But other researchers are spotting on important concerns about academic honesty, of students becoming over AI dependent, other concern is about the possible reduction of critical thinking skills (Chan and Hu, 2023; Ng et al., 2025). A closer look at the research shows something important: AI seems to handle lower-level thinking tasks like summarizing or translating very well. But when it comes to higher order thinking, analyzing, evaluating, creating the outcome depends on how the teaching is designed (Ibrahim et al.

This brings us to the central focus of the present study. Rather than assuming that the use of generative AI automatically leads to improved learning or responsible use, this study examines how students' AI literacy, perceived learning benefits, and ethical awareness are related following participation in a structured, activity-based learning experience. The study is particularly interested in whether students who demonstrate greater AI literacy

also perceive greater learning benefits and show greater awareness of ethical issues associated with AI use. It further investigates whether perceived learning benefits may serve as a mechanism linking AI literacy to ethical awareness. In addition, the study examines whether these outcomes differ according to students' gender, prior AI experience, and frequency of AI use. Drawing on constructivist learning perspectives and established frameworks of AI literacy (Kong et al., 2021; Long and Magerko, 2020), the study therefore explores whether structured engagement with generative AI can support more active, critical, and responsible forms of student participation in AI-assisted learning.

The present study

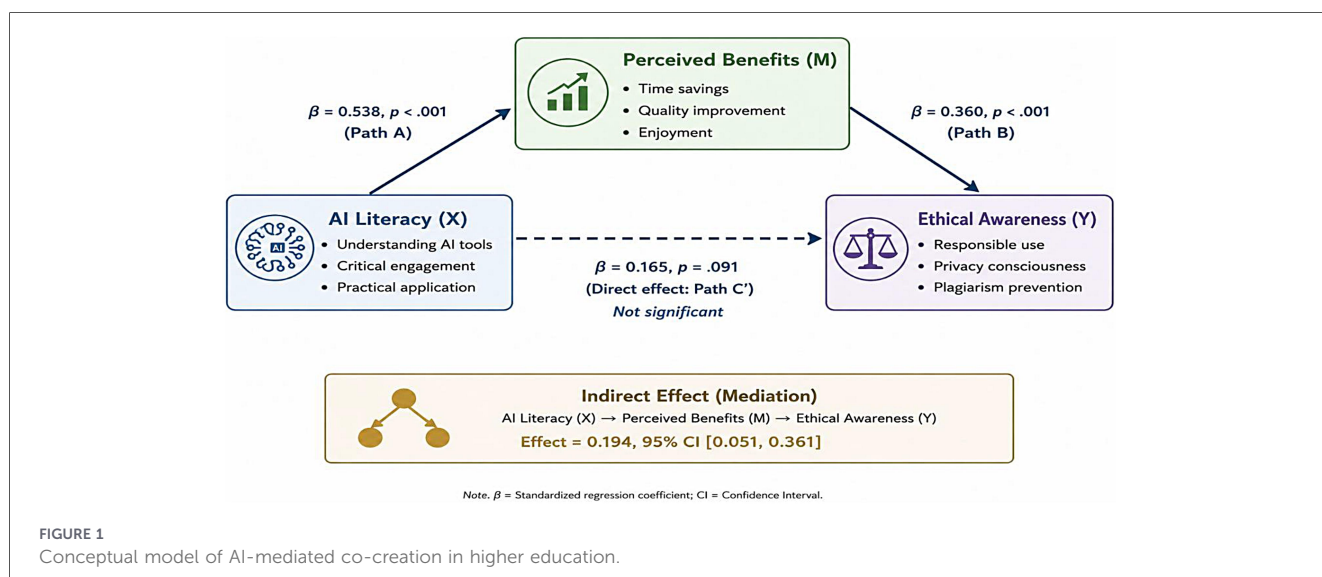
This study looks at a two-semester classroom activity. We asked undergraduate students to prepare and deliver short lessons to their peers—but with a twist. They had to use generative AI tools like ChatGPT, Google Gemini, and Microsoft Copilot to help them. After the lessons, we sat down together for guided, reflective discussions. We talked about the tools they used, what kind of outputs the AI gave them, and the ethical questions that come up when students use AI in academic settings.

The whole activity was built around one core idea: we wanted students to be co-creators of knowledge, not just passive consumers. That meant they had to do more than just copy-paste from the AI. They had to think critically about what the AI produced, catch and fix its mistakes, and use their own judgment when explaining things to their classmates.

With this design in mind, we let the research questions guide our study.

Significance of the study

This study addresses several important gaps in existing research. First, while much of the research on AI in education is still conceptual or theoretical, this study provides real evidence from a structured, long-term classroom activity with 104



undergraduate students. Second, “Few empirical studies have examined AI literacy, perceived learning benefits, and ethical awareness within the same structured AI-integrated classroom activity. Third, the study examines the co-creation concept as a theoretical framework for understanding students’ active engagement with AI-generated content rather than as an independently measured construct.

By addressing these gaps, this study adds to the expanding real-world evidence on AI-integrated teaching and gives educators solid ideas for adopting critical, responsible, and creative engagement with generative AI tools.

Literature review

Generative AI in higher education

Generative AI has become central to higher education, offering benefits such as personalized learning, instant feedback, and idea generation (Bansal et al., 2024; Michel-Villarreal et al., 2023; Pesovski et al., 2024; Rusmiyanto et al., 2023; Shanto et al., 2025). However, concerns about academic integrity, over-reliance, and reduced critical thinking persist (Chan and Hu, 2023; Ng et al., 2025). Ibrahim et al. (2023) found that ChatGPT performed well on lower-level tasks but showed no clear advantage in higher-order thinking, emphasizing that student performance depends on instructional design.

Chan and Hu (2023) reported that students value AI’s support but worry about losing critical thinking capacity. This highlights the risk of passive AI consumption without structured pedagogical design. The present study addresses this by requiring students to identify and correct AI errors, fostering critical evaluation skills.

Critical thinking in AI-mediated learning

Critical thinking—the ability to analyze, evaluate, and synthesize information—is essential for 21st-century learners (Facione, 2020). Research presents two contrasting views: Lee (2024) suggests AI can strengthen critical thinking through iterative questioning, while Urhan et al. (2026) warns that over-reliance may weaken higher-order thinking. Gonsalves (2024) frames AI as either a “cognitive amplifier” or “cognitive crutch,” depending on instructional design (Oschinsky and Nguyen, n.d.).

Empirical evidence is mixed. De la Puente et al. (2024) found that ChatGPT-enhanced debates improved critical thinking. Conversely, Urhan et al. (2026) found that ChatGPT’s reasoning is sometimes flawed, with weaker students accepting invalid claims uncritically. Vazquez-Cano et al. (2023) described ChatGPT as “the brightest student in the class,” warning that this may discourage student engagement if not paired with co-creative teaching approaches.

AI literacy: conceptual foundations

AI literacy involves understanding, evaluating, and using AI tools effectively and responsibly (Kong et al., 2021; Long and

Magerko, 2020; Ng et al., 2021). Tadimalla and Maher (2025) outline four pillars: technical understanding, practical interaction, critical/ethical evaluation, and societal awareness. Farrelly and Baker (2023) map AI literacy onto Bloom’s Taxonomy, emphasizing evaluation and creation. Petrova (2026) found expert consensus that universal AI literacy (4.8/5) and bias audits (4.5/5) are essential. Amdan et al. (2026) warn that excessive AI reliance reduces critical thinking, reinforcing the need for balanced literacy.

Student perceptions and the risk of over-reliance

Students generally view AI positively for personalized learning, writing, and research (Chan and Hu, 2023). However, they express concerns about over-reliance, with one student stating, “I can become over-reliant on generative AI technologies” ($M = 2.85$). Delcker et al. (2024) found that attitude toward AI ($\beta = 0.26$, $p < .01$) predicts usage, while actual skills had a negative effect ($\beta = -0.10$). Hoernig et al. (2024) warn that dependency may lead to reduced critical thinking, creativity, and social skills.

Ethical awareness and responsible AI use

Reina Marin et al. (2025) found that 61.1% of faculty and 53.5% of students perceive AI systems as insufficiently transparent. They stress that “automation should not replace critical thinking or human participation.” An et al. (2024) proposed an ethics guideline emphasizing human autonomy, agency, and academic integrity. Arslan (2026), (Corlee, 2025) identified teacher concerns about privacy, data governance, and reduced student responsibility, warning that “if control is missing, students may be misguided.” Perera and Lankathilaka (2023) summarize risks including reduced creativity, critical thinking, and ethical issues like bias and plagiarism (Spivakovsky et al., 2023; Tsao, 2025).

Constructivist pedagogy and co-creation

The intervention is grounded in constructivist theory (Vygotsky, 1978), emphasizing active, social, and reflective learning. Vygotsky’s Zone of Proximal Development (ZPD) positions AI as a cognitive scaffold that supports but does not replace thinking. Schön’s (1983) “reflective practitioner” concept aligns with students questioning AI outputs. The “protégé effect” (Chase et al., 2009) suggests that teaching others deepens understanding. Vygotsky (1978) and Chase et al. (2009) emphasize maintaining human judgment, positioning AI as a partner, not a replacement. These principles guided the study’s design, promoting co-creation over passive consumption.

The mediating role of perceived benefits in AI-mediated learning”

Research on technology acceptance has consistently demonstrated that perceived usefulness and perceived benefits

are critical determinants of sustained technology engagement (Davis, 1989; Venkatesh et al., 2003). In educational contexts, students who perceive that technology enhances their learning outcomes are more likely to adopt responsible usage patterns (Scherer et al., 2019).

Self-Determination Theory (Ryan and Deci, 2000) provides additional insight: when students experience competence (feeling capable of using AI effectively) and autonomy (feeling in control of AI use), they are more likely to internalize responsible behaviors. Perceived learning benefits serve as evidence of competence, reinforcing the intrinsic motivation to use AI ethically.

Recent empirical work supports this mediation pathway. Delcker et al. (2024) found that positive attitudes toward AI ($\beta = 0.26$, $p < .01$) predicted usage, suggesting that perceived value drives engagement. Similarly, Mirea et al. (2025) reported that students who perceived AI as beneficial were more likely to express concerns about ethical use, implying that benefits awareness precedes ethical reflection.

Research gap

There's plenty of interest in generative AI and critical thinking these days, but the research itself has some noticeable gaps.

First: A lot of what's out there isn't based on real classroom data it's conceptual (Noroozi et al., 2024). And if you read the systematic reviews, they keep saying the same thing: we need more empirical studies, actual studies in classrooms, looking at how AI affects higher-order thinking (Batista et al., 2024; Weng et al., 2024).

Second: Most studies treat AI as basically a text generator or a summarizer. You don't see many structured interventions where students have to be co-creators—where they actually have to evaluate what the AI gives them and correct it. Ng et al., 2025 makes that point.

Third: people have studied AI literacy (Kong et al., 2021), critical thinking (de la Puente et al., 2024), and ethical awareness (Reina Marin et al., 2025) separately. But almost no one has looked at all three in the same study. Chan and Hu (2023) mention this.

Fourth: and this is a problem—there's barely any empirical research from Middle Eastern higher education (Khlouf, 2025). Like, almost none.

Fifth: Not many studies even ask whether these AI activities work equally well for different gender groups. UNESCO (2019) flagged this years ago.

Theoretical framework

Overview

This framework synthesizes five interconnected domains—AI literacy, perceived learning benefits, critical thinking, co-creation, and ethical awareness—into a coherent model explaining how structured generative AI activities shift students from passive consumers to active co-creators. Core Theoretical Constructs

AI literacy as a foundational competency

AI literacy serves as the foundational competency that enables students to understand, evaluate, and use AI tools effectively and responsibly (Kong et al., 2021; Long and Magerko, 2020). Drawing on Tadimalla and Maher's four-pillar framework (2025) and Petrova's validated themes (2026), we conceptualize AI literacy as encompassing four interrelated dimensions:

- Technical understanding: knowledge of how AI tools function, including their capabilities and limitations
- Critical evaluation: the ability to judge AI outputs for accuracy, relevance, and potential bias
- Ethical judgment: awareness of issues related to privacy, plagiarism, and responsible use
- Practical application: the capacity to use AI tools effectively for specific academic tasks

In the present study, AI literacy was operationalized as students' perceived ability to understand how AI can support academic tasks and to engage critically with AI-generated content within the context of the classroom activity. Three Likert-scale items were used to capture these perceptions. The scale demonstrated acceptable internal consistency (Cronbach's $\alpha = .751$).

Critical thinking as a mediating process

Drawing on Facione (2020) and the empirical work of de la Puente et al. (2024) and Urhan et al. (2026), we acknowledge that critical thinking is not directly measured as a variable in this study. Rather, it informs the pedagogical design of the instructional activity and serves as a theoretical lens for interpreting the findings. The structured activity—requiring students to prepare lessons, evaluate AI-generated content, and engage in reflective discussions—was explicitly designed to activate critical thinking processes, including analysis, evaluation, synthesis, and reflection (Facione, 2020; Stanford, 2025).

However, the mediating mechanism tested in our empirical model is perceived learning benefits, not critical thinking itself. Drawing on Self-Determination Theory (Ryan and Deci, 2000) and Expectancy-Value Theory (Eccles and Wigfield, 2002), we propose that the relationship between AI literacy and ethical awareness operates through students' perceptions of the tangible benefits derived from AI use (e.g., time savings, improved quality, enhanced enjoyment). When students develop AI literacy, they become better able to recognize these benefits. It is argued that such perception of value proposed to be linked to ethical awareness in that students valuing what they gain from AI are likely to adopt ethical attitudes towards their usage of AI. Such relationship between the two constructs is tested empirically in the subsequent parts of the paper. In learning environments where AI is involved, critical thinking processes remain essential. As Ng et al., 2025 distinguishes, when students engage critically with AI, it works as a cognitive amplifier that enhances thinking; but when they consume AI passively, it

becomes a cognitive crutch that does the thinking for them (Stanković et al., 2025). Our instructional design was therefore structured to foster these critical thinking processes, ensuring that students engaged actively with AI outputs rather than accepting them uncritically.

The relationship between these concepts can be understood as follows:

Thus, while critical thinking underpins the pedagogical design, perceived benefits function as the empirically tested mediator in our conceptual model.

Co-creation as an aspirational framework

Co-creation is conceptualized in this study as a theoretical description of active AI-mediated learning, representing a shift from passive consumption toward active knowledge production. Co-creation was not operationalized or measured as an independent construct in the questionnaire. Therefore, references to co-creation in this study describe the instructional orientation of the activity rather than an empirically tested outcome.

Drawing on Vygotsky (1978) and Chase et al. (2009), co-creation in AI-mediated learning involves:

- Intellectual ownership: taking responsibility for AI-enhanced work
- Creative synthesis: generating novel outputs that integrate AI-generated and human-generated elements
- Metacognitive control: exercising deliberate oversight over AI use
- Peer teaching: translating AI-assisted understanding into pedagogically effective explanations for others

Although co-creation was not operationalized as a measured variable, it serves as the aspirational framework guiding the study's instructional design and interpretation. This conceptualization directly operationalizes the “create” level of Bloom's taxonomy as applied to AI literacy by Farrelly and Baker (2023) and Zhou and Schofield (2024).

Ethical awareness as interdependent outcome

Drawing on Reina Marin et al. (2025), An et al. (2024), and UNESCO (2019), ethical awareness is conceptualized in two interrelated ways: first, as an integral component of AI literacy (reflected in the “ethical judgment” pillar); and second, as an outcome that emerges from co-creative engagement with AI tools.

Ethical awareness encompasses four dimensions:

- Privacy consciousness: understanding how data is collected and used
- Plagiarism prevention: knowing when to give credit and how to follow academic honesty rules
- Bias detection: recognizing when AI outputs may be biased or discriminatory
- Responsible use commitment: intending to use AI tools in an honest and transparent manner

Proposed conceptual model

Based on the theoretical synthesis above, the following conceptual model is proposed:

Theoretical propositions

From the conceptual model outlined above, we offer the following propositions:

1. Structured, task-based AI activities—where students evaluate, correct, and take ownership of AI-generated content—improve AI literacy more effectively than unstructured AI use.
2. AI literacy may support students' critical evaluation of AI-generated content. However, because critical thinking was not independently measured as a separate construct in the present study, this proposition is presented as a theoretical interpretation rather than an empirically tested relationship.
3. Perceived learning benefits mediate the relationship between AI literacy and ethical awareness. Students who perceive greater benefits from AI use (e.g., efficiency, quality enhancement, enjoyment) are more likely to develop a sense of responsibility and commitment to ethical AI use, as they recognize that responsible use sustains these benefits.
4. The relationship between AI literacy and ethical awareness is statistically associated with perceived learning benefits through a significant indirect (mediated) pathway. This is consistent with the proposition that AI literacy alone may not be sufficient to explain variation in ethical awareness; perceived benefits appear to be a relevant intervening variable.
5. Ethical awareness requires deliberate cultivation through reflective discussions and structured activities that surface ethical issues and make them teachable.
6. Well-designed AI activities with appropriate pedagogical scaffolding can produce equitable outcomes across gender groups.

Alignment with empirical findings

The empirical findings provide support for the relationships among the three measured constructs: AI literacy, perceived learning benefits, and ethical awareness. The observed correlations were statistically significant ($r = 0.559$, $p < .001$; $r = 0.386$, $p < .001$; and $r = 0.472$, $p < .001$, respectively).

Mediation analysis indicated a statistically indirect association between AI literacy and ethical awareness through perceived learning benefits (Indirect effect = 0.194, 95% CI [0.051, 0.361]), with a non-significant direct effect ($\beta = 0.165$, $p = .091$). These findings support the empirically tested relationships among the measured variables, while critical thinking and co-creation remain theoretical components of the conceptual framework.

Summary

The theoretical framework integrates AI literacy, critical thinking, co-creation, perceived learning benefits, and ethical awareness as complementary concepts in AI-mediated learning. The empirical investigation focuses specifically on the three measured constructs: AI literacy, perceived learning benefits, and ethical awareness.

Critical thinking and co-creation provide the theoretical and instructional context for understanding the structured activity but were not independently measured or statistically tested. The key theoretical contribution is the identification of perceived learning benefits as a statistically significant mediating (intervening) variable linking AI literacy to ethical awareness, grounded in Self-Determination Theory (Ryan and Deci, 2000) and Expectancy-Value Theory (Eccles and Wigfield, 2002).

Research design & methodology

This study adopts a post-intervention explanatory design to investigate the relationships among the study variables and to test theoretically derived hypotheses. Data were collected after students completed a structured, semester-long AI-integrated classroom activity. A deductive approach was applied, where hypotheses were formulated based on existing literature and then tested using observed data.

The study employs a quantitative survey design as the primary method of data collection and analysis. Quantitative data were analyzed using SPSS. Descriptive statistics were used to summarize participant responses, while Cronbach's alpha was employed to assess internal consistency reliability. Pearson correlation analysis was conducted to examine relationships among variables, followed by mediation analysis using Hayes' PROCESS Macro (Model 4) with 5,000 bootstrap samples to investigate indirect effects. Non-parametric tests (Mann-Whitney U and Kruskal-Wallis) were applied where appropriate due to non-normal data distributions (Shapiro-Wilk, $p < .05$).

This design was chosen because it allows for the systematic examination of associations among the key constructs and enables testing of the proposed mediation model while acknowledging the cross-sectional nature of the data.

Participants

The study targeted students enrolled in the academic year 2024–2025 who participated in an AI-integrated classroom activity over two academic semesters (Fall and Spring). Of 300 students invited, 104 completed the survey voluntarily (response rate = 34.6%).

Data collection

This questionnaire was administered after completing the instructional task. It included sections pertaining to

demographics, the use of AI, benefits, ethics, problems, and perceptions, using the five-point Likert scale for recording the responses. Even though there were some open-ended questions in the questionnaire, their analysis is not considered systematic and has therefore not been included in the findings. Hence, the current study can be classified as a quantitative, cross-sectional, post-activity study.

Study domains and approvals

In this study, an educational activity in a classroom setting was conducted, after which a questionnaire survey was carried out among 104 adult subjects. Ethical board or Research Ethics Committee approval or waiver of ethical approval was obtained prospectively prior to data collection.

Participation in the research process was completely voluntary. Participants were notified about the fact that their decision whether to take part or refuse from it would not affect their evaluation, grades or relationship with the researcher. In order to avoid the possible influence of the power differential between the instructor and students, the research component was entirely separated from academic process and data collection happened after the completion of the final exams.

All the participants received clear explanations regarding the aim, procedures and nature of the study before the start of their involvement. Participants were informed about their right to refuse participation without any negative consequences. All appropriate actions were taken in order to protect the privacy and anonymity of the participants. There was no directly identifying information in the analytic dataset and collected data were used exclusively for research purposes.

There were no vulnerable participants in this study, and there were no clinical, biological, or invasive interventions. The researchers considered this activity and survey to represent low-risk educational research. There were no conflicts of interest identified by the researchers.

Survey instrument

The questionnaire was developed in alignment with the study objectives and included multiple-choice and Likert-scale items. The instrument assessed students' perceived AI literacy in the context of classroom activity, perceived learning benefits, ethical awareness, challenges, and overall perceptions. The AI-literacy items focused on students' perceived understanding of AI-assisted academic use and their critical engagement with AI-generated content.

Data analysis

Quantitative data were analyzed using SPSS. Descriptive statistics were used to summarize participant responses, while Cronbach's alpha was employed to assess internal consistency reliability. Pearson correlation analysis was conducted to examine relationships among variables, followed by Mediation analysis to investigate indirect effects among variables.

In addition, non-parametric tests were applied where appropriate. The Mann–Whitney U test was used to examine differences between two independent groups, while the Kruskal–Wallis test was used to assess differences across multiple groups when applicable.

Research questions

- RQ1: What are students' levels of AI literacy, perceived learning benefits, and ethical awareness following participation in a structured AI-integrated classroom activity?
- RQ2: What are the relationships among AI literacy, perceived learning benefits, and ethical awareness among students following participation in the activity?
- RQ3: Does perceived learning benefits mediate the relationship between AI literacy and ethical awareness among students?
- RQ4: Are there are significant differences in AI literacy, perceived learning benefits, and ethical awareness based on gender, prior AI experience, and frequency of AI use?

Hypotheses

- H1: AI literacy has a significant positive effect on perceived learning benefits.
- H2: Perceived learning benefits have a significant positive effect on ethical awareness.
- H3: AI literacy has a significant positive direct effect on ethical awareness, after accounting for perceived learning benefits.
- H4: Perceived learning benefits significantly mediate (indirect effect) the relationship between AI literacy and ethical awareness.
- H5: There are significant differences in AI literacy, perceived learning benefits, and ethical awareness based on gender.
- H6: There are significant differences in AI literacy, perceived learning benefits, and ethical awareness based on prior AI experience.
- H7: There are significant differences in AI literacy, perceived learning benefits, and ethical awareness based on frequency of AI use.

Testing reliability and validity:

The internal consistency of the survey instrument was assessed using Cronbach's Alpha coefficients (Taherdoost, 2016). As shown in Table 1, all subscales exceeded the

acceptable threshold of 0.70. Perceived Benefits demonstrated the highest reliability ($\alpha = 0.894$), followed by Ethical Awareness ($\alpha = 0.800$), and AI Literacy ($\alpha = 0.751$). The overall instrument reliability was $\alpha = 0.861$.

The reliability and the association between the items were determined by corrected item-total correlation. The item-total correlations were found to be acceptable to high ($p < .01$) for all of the items. The AI Literacy item correlations varied between 0.521 and 0.686, the Perceived Benefits item correlations between 0.772 and 0.808, and the Ethical Awareness item correlations between 0.491 and 0.740. The construct validity of the instrument was not assessed independently and it requires further investigation.

Descriptive analysis

Demographics

Table 2 presents the demographic data of the study participants ($N = 104$). Regarding gender, the sample was (50%) were male and (50%) female respondents, for each group.

In terms of age, most respondents (56.7%) were (21–23) age range, which aligns with the targeted sample. and supported by the academic level data, which indicates that (95.2%) of the participants were in bachelor's programs (BCs), while postgraduate students were (4.8%).

Background/experience & outcomes variables

Data in Table 3 shows a high level of AI familiarity among participants, as (91.3%) have prior usage of AI tools. These participants also showed high engagement levels, with (56.7%) using AI either "Often" or "Very frequently." On the other hand, only a small fraction of the sample (8.7%) had no previous experience. This high rate indicates that the respondents possess sufficient practical background to provide informed perspectives on the study's constructs. Regarding the outcomes, participants reported notable improvements following the activity. Specifically, (65.4%) stated that the activity improved their learning quality "a lot," while (32.7%) said it improved "somewhat," with only (1.9%) reporting no improvement. Similarly, when asked about the overall quality of their learning, (65.4%) responded "Yes, definitely," and the remaining (34.6%) responded "Yes, to some extent," with no participants indicating a lack of improvement.

TABLE 1 Survey instrument reliability & validity.

Reliability			Validity			
Subscales	Cronbach's alpha coefficient	N	1	2	3	4
AI literacy	0.751	3	.521**	.686**	.549**	
Perceived benefits	0.894	3	.772**	.808**	.800**	
Ethical awareness	0.800	4	.567**	.491**	.740**	.680**
Overall reliability	0.861	10				

**Denotes statistical significance at $p < .01$ (correlation is significant at the 0.01 level, 2-tailed).

AI Adoption, Applications, and Perceived Impact:

Data showed alignment between the AI tools used during the activity and the specific areas where participants seek assistance. The data reveals that ChatGPT and Google Gemini serve as the

TABLE 2 Demographic data (personal data).

Variable	Category	Frequency	Percent
Gender	Female	52	50.0%
	Male	52	50.0%
Age	18–20	33	31.7%
	21–23	59	56.7%
	24+	12	11.5%
Academic level	BCs	99	95.2%
	Postgraduate	5	4.8%
	Total	104	100%

TABLE 3 Background/experience & outcomes.

Variable	Response	Frequency	Percent
Have you used AI tools before this activity?	Yes	95	91.3%
	No	9	8.7%
If yes, how frequently?	Rarely	9	8.7%
	Sometimes	32	30.8%
	Often	31	29.8%
	Very frequently	28	26.9%
Do you think this activity improved the Learning quality of your learning?	Yes, a lot	68	65.4
	Yes, somewhat	34	32.7
	No	2	1.9
	improvement		
Do you think this activity improved the overall quality of your learning?	Yes, definitely	68	65.4
	Yes, to some extent	36	34.6
	No	0	0
	Total	104	100%

primary “all-in-one” solutions, effectively covering all academic needs, with a particular concentration in Summarizing lessons and Brainstorming ideas.

Notably, despite the availability of specialized tools coding or presentations for example, students showed a significant reliance on general-purpose Large Language Models (LLMs) to solve complex tasks like problem-solving and translation. This shows that students prefer using AI tools that can do many things rather than tools designed for one specific task. This means students like having one main AI tool to help them with different academic tasks.

Perceived benefits, skill enhancement, and experienced challenges

Using AI tools in learning has brought clear academic benefits. Most students (81.7%) said they understood the topic better, and) 71.2% (said AI helped them save time. These benefits are linked to better skills. Students reported improvements in problem-solving (71.2%), research skills) 70.2%), and critical thinking (62.5%). This suggests that AI helps students think more deeply about difficult academic content.

However, using AI for learning also comes with some problems, as shown in Figure 2. The main concern was that AI outputs can be wrong or misleading (54.8%). Next, 35.0% of students worried about plagiarism. Other issues included lack of training (32.0%) and poor internet connection (31.1%).

The results show that students like how AI saves time and effort, but they also know its limits. This means AI should be used as a helpful tool, not as the main source of information. Schools should focus on teaching digital skills and creating clear rules for using AI ethically.

Study domains

Chi-square goodness-of-fit test

To assess whether the distribution of responses across the study domains differed significantly from an equal distribution, a Chi-Square Goodness-of-Fit test was conducted for the five-

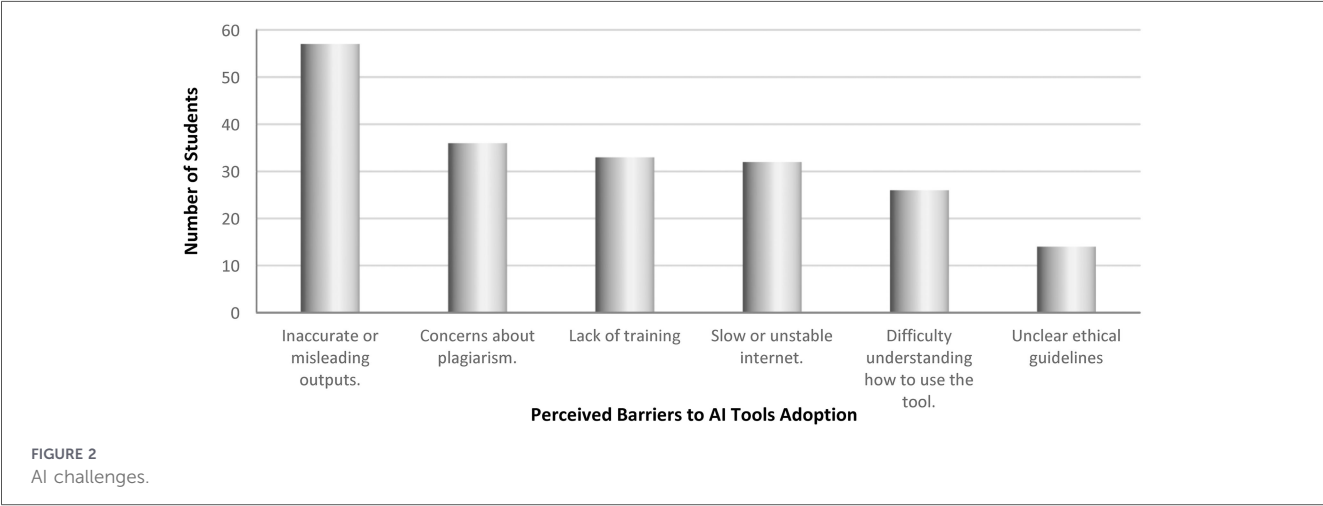


TABLE 4 Study domains.

Domain 1 AI literacy						
Item	W.A	S.D	Rank	Level	Chi ² (GOF)	P-value
The activity helped me understand how AI can assist in lesson preparation.	4.01	1.05	1	High	64.462	<i>p</i> < .001
The activity encouraged me to think critically about AI-generated content.	3.84	1.04	2	High	47.346	<i>p</i> < .001
I was actively engaged during the AI-based activity	3.56	1.22	3	High	23.404	<i>p</i> < .001
Total	3.81	0.908		High	45.308	<i>p</i> < .001
Domain 2 perceived benefits						
AI helped me save time in lesson preparation.	4.269	0.927	2	Very high	94.077a	<i>p</i> < .001
AI improved the quality of the content I produced.	4.087	1.025	3	High	71.673a	<i>p</i> < .001
AI made the learning process easier and more enjoyable.	4.288	0.931	1	Very high	101.000a	<i>p</i> < .001
Total	4.215	0.874		Very high	118.115b	<i>p</i> < .001
Domain 3 ethical awareness						
I have become more aware of the ethical issues related to AI use.	3.885	1.109	4	High	45.615a	<i>p</i> < .001
I believe clear guidelines should be established for AI use in education.	4.067	1.134	3	High	67.442a	<i>p</i> < .001
I am committed to using AI responsibly and ethically.	4.279	0.990	1	Very high	101.577a	<i>p</i> < .001
After the activity, I better understand how AI can be used ethically.	4.183	1.041	2	High	86.769a	<i>p</i> < .001
Total	4.103	0.846		High	93.885c	<i>p</i> < .001

TABLE 5 Group comparisons across demographic variables.

Domain		AI literacy	Perceived benefits	Ethical awareness
Gender	Mann–Whitney	821.500	881.500	925.000
	<i>p</i> -value	0.212	0.435	0.690
Academic level	Mann–Whitney	200.500	226.500	177.500
	<i>p</i> -value	0.472	0.741	0.281
Prior experience in AI	Mann–Whitney	270.500	179.000	186.000
	<i>p</i> -value	0.067	0.003	0.005
Age	Kruskal Wallis	1.859	1.026	1.032
	<i>P</i> -value	0.395	0.599	0.597
Using AI tools frequency	Kruskal Wallis	12.013	12.068	11.406
	<i>P</i> -value	0.007	0.007	0.010
Subject understanding improvement	Kruskal Wallis	7.777	14.235	8.239
	<i>P</i> -value	0.020	0.001	0.016
Overall improvement	Kruskal Wallis	11.648	8.095	3.416
	<i>P</i> -value	0.001	0.004	0.065

point Likert-scale responses. Under the null hypothesis, responses were expected to be equally distributed across the five response categories, with an expected proportion of 20% for each category. The results indicated that the observed response distributions differed significantly from an equal distribution across the five response categories.

The weighted average (W.A.) was reported as a descriptive summary measure of central tendency; however, it was not used as the variable tested by the Chi-Square statistic. The Chi-Square test was based on the observed frequencies of responses within the five Likert-scale categories.

Directional confirmation: Wilcoxon signed-rank tests

To determine whether students’ responses were significantly above the neutral midpoint (3.0 on a 5-point scale), one-sample Wilcoxon signed-rank tests were conducted. Results showed that the median scores for all three domains were significantly higher than 3.0:

These results confirm that students’ responses were not only non-uniformly distributed (as indicated by the Chi-Square tests) but also directionally positive, demonstrating clear consensus above the neutral midpoint.”

TABLE 6 Mediation analysis results.

Path	β	SE	<i>t</i>	<i>p</i>	LLCI	ULCI
Direct effects						
AI literacy → Perceived benefits	0.538	0.079	6.810	<.001	0.381	0.695
Perceived benefits → Ethical awareness	0.360	0.101	3.573	<.001	0.160	0.561
AI literacy → Ethical awareness	0.165	0.097	1.704	0.091	−0.027	0.358
Indirect effect	Effect	BootSE			BootLLCI	BootULCI
AI literacy → Perceived benefits → Ethical awareness	0.194	0.080			0.051	0.361

TABLE 7 Summary of hypothesis testing.

Hypothesis	Statement	Decision	Evidence
H1	AI literacy has a significant positive effect on perceived learning benefits.	Supported	$\beta = 0.538, p < .001$
H2	Perceived learning benefits have a significant positive effect on ethical awareness.	Supported	$\beta = 0.360, p < .001$
H3	AI literacy has a significant positive direct effect on ethical awareness.	Rejected	$\beta = 0.165, p = .091$
H4	Perceived learning benefits mediate the relationship between AI literacy and ethical awareness.	Supported	Indirect effect = 0.194, 95% CI [0.051, 0.361]
H5	Significant gender differences exist across study domains.	Rejected	$p > .05$ for all domains
H6	Significant differences exist based on prior AI experience.	Partially Supported	Significant for Perceived Benefits ($p = .003$) and Ethical Awareness ($p = .005$); not for AI Literacy ($p = .067$)
H7	Significant differences exist based on frequency of AI use.	Supported	$p < .05$ for all domains

Domain results: AI literacy, perceived benefits, and ethical awareness

As presented in Table 4, participants reported high levels across all three domains. AI Literacy scored (W.A = 3.81,

SD = 0.908), with the highest-rated item being understanding how AI assists in lesson preparation (W.A = 4.01), followed by critical thinking about AI-generated content (W.A = 3.84), and active engagement (W.A = 3.56).

Perceived Benefits scored very high (W.A = 4.215, SD = 0.874). Students rated “AI made learning easier and more enjoyable” highest (W.A = 4.288), followed by time savings (W.A = 4.269), and improved quality of work (W.A = 4.087).

Ethical Awareness scored high (W.A = 4.103, SD = 0.846). The highest-rated item was commitment to responsible AI use (W.A = 4.279), followed by understanding ethical use after the activity (W.A = 4.183), and support for institutional guidelines (W.A = 4.067). Chi-square Goodness-of-Fit tests confirmed significant deviations from uniform distribution across all domains ($p < .001$). Combined with the one-sample Wilcoxon signed-rank test results reported above, these findings indicate that students demonstrated a positive directional consensus in their responses, consistently above the neutral midpoint.

Inferential analysis: demographic differences

Non-parametric tests (Mann–Whitney U and Kruskal–Wallis) were used due to non-normal data distributions (Shapiro–Wilk, $p < .05$). Table 4 summarizes the results. No significant differences were found for Gender or Academic Level across any domain ($p > .05$). Prior AI Experience significantly differentiated Perceived Benefits ($p = .003$) and Ethical Awareness ($p = .005$). Age had no significant effects. Frequency of AI Usage and Subject Understanding Improvement showed significant differences across all domains ($p < .05$), while Overall Improvement was significant for AI Literacy ($p = .001$) and Perceived Benefits ($p = .004$), but not Ethical Awareness ($p = .065$).

Mediation analysis results

For the mediation analysis, a bootstrapping procedure (5,000 samples) was used, as it is robust against violations of normality assumptions. To examine the mediating role of Perceived Benefits (M) in the association between AI Literacy (X) and Ethical Awareness (Y), Hayes’ PROCESS Macro (Model 4) was

utilized with 5,000 bootstrap samples. The results revealed the following:

Path A ($X \rightarrow M$): AI Literacy was significantly and positively associated with Perceived Benefits ($\beta = 0.538$, $p < .001$), explaining 31.2% of its variance.

Path B ($M \rightarrow Y$): Perceived Benefits was significantly and positively associated with Ethical Awareness ($\beta = 0.360$, $p < .001$), after controlling for AI Literacy.

Direct association ($X \rightarrow Y$): The direct association between AI Literacy and Ethical Awareness was not statistically significant after accounting for Perceived Benefits ($\beta = 0.165$, $p = .091$).

Indirect effect ($X \rightarrow M \rightarrow Y$): The indirect effect was statistically significant (Effect = 0.194, BootLLCI = 0.051, BootULCI = 0.361). Because the bootstrap confidence interval does not include zero, the results provide evidence consistent with an indirect mediation pathway.

Conclusion

The results indicate a statistically significant indirect association between AI Literacy and Ethical Awareness through Perceived Benefits, while the direct association between AI Literacy and Ethical Awareness was not statistically significant. These findings are consistent with the interpretation that the association between AI Literacy and Ethical Awareness may operate indirectly through Perceived Benefits. However, given the cross-sectional design, these findings are correlational and should not be interpreted as evidence of a causal pathway. Higher levels of perceived AI Literacy were associated with higher Perceived Benefits, which were in turn associated with higher Ethical Awareness.

Summary of hypothesis testing

Discussion

The aim of this study was to analyze the influence of the class activity based on the application of generative AI on the AI literacy, learning benefits perceived by undergraduate students, and their ethical awareness. The findings have shown a statistically significant indirect effect of the AI literacy on the ethical awareness via the perception of learning benefits, as hypothesized in the proposed theory. This relationship is in line with—but does not prove—the possible process through which AI literacy is associated with the perception of the usefulness of AI, which is further associated with perceived learning benefits (savings in time, improvement in quality, pleasure) and, finally, with ethical awareness. This finding aligns with Self-Determination Theory (Ryan and Deci, 2000). When students experience competence through perceived benefits, they are more likely to internalize responsible behaviors. It also supports Expectancy-Value Theory (Eccles and Wigfield, 2002), where value beliefs (perceived benefits) mediate the relationship between competence beliefs (AI literacy) and behavioral outcomes (ethical awareness).

Notably, the insignificant direct effect ($\beta = 0.165$, $p = .091$) together with the significant indirect effect suggests that AI

literacy does not have a direct association with ethical awareness when learning gains are controlled for. This implies several important implications for teaching practice: instructors must not rely on the belief that the mere teaching of AI techniques fosters responsible conduct. Students must also experience and value the benefits of AI use.

Summary of key findings

The results showed that after the AI activity, students had good AI literacy ($M = 3.81$), very good perceived benefits ($M = 4.22$), and good ethical awareness ($M = 4.10$). Students said that AI helped them save time (71.2%), improved their ability to solve problems (71.2%), and strengthened their critical thinking (62.5%). At the same time, some students worried about incorrect AI outputs (54.8%) and plagiarism (35.0%). This shows that students have a balanced and thoughtful view of AI tools.

The statistical analysis also showed that students who had used AI before the activity had higher perceived benefits ($p = .003$) and higher ethical awareness ($p = .005$). However, gender and academic level did not make any difference in the results. Also, how often students used AI was linked to all three areas ($p < .01$). This means that using AI more often helps students build better skills and awareness.

Mediation analysis

The mediation analysis indicated a statistically significant indirect association between AI literacy and ethical awareness through perceived learning benefits. The direct association between AI literacy and ethical awareness was not statistically significant after accounting for perceived learning benefits. However, because the study used cross-sectional data, these findings should be interpreted as evidence of a statistical indirect association rather than as evidence of a causal or temporal mediation process.

The findings indicate that higher AI literacy was associated with higher perceived learning benefits, and higher perceived learning benefits were in turn associated with higher ethical awareness. These associations are consistent with the proposed mediation model, but they should not be interpreted as evidence that AI literacy causes changes in perceived benefits or ethical awareness; rather, AI literacy appears to be associated with greater perceived benefits, and these perceived benefits, in turn, are associated with higher ethical awareness. It is important to note, however, that these associations are correlational and should not be interpreted as causal pathways. A summary of all hypothesis testing outcomes is presented in Table 7.

Interpretation within theoretical framework

This finding is in line with constructivist learning theories (Vygotsky, 1978) and the protégé effect (Chase et al., 2009). In

order to accomplish this task, the participants had to use and teach with AI applications, which entailed an active participation with the material, evaluation of AI products and teaching decisions. Such an active and social learning experience might reasonably be connected to the identified correlations between literacy and ethical reasoning, but the cross-sectional nature of the study precludes any firm conclusions on causality. Furthermore, the results support the distinction proposed in the introduction between passive consumption and active co-creation. Students who used AI as a cognitive scaffold rather than a substitute for thinking reported higher levels of both perceived benefits and ethical awareness.

This suggests that the structured, guided nature of the activity was essential in shifting students' orientation from consumers to co-creators.

Comparison with previous literature

In addition to the results of the earlier studies, the results of this research establish that the perception of benefits is a mediator in the link between literacy and ethics. While earlier research has individually highlighted the influence of artificial intelligence in enhancing the learning process (Bansal et al., 2024; Rusmiyanto et al., 2023) and the relevance of ethical awareness (An et al., 2024; Reina Marin et al., 2025), the current research sheds light on the interaction among the mentioned constructs. The mediation model provides a statistical account of the associations among these constructs: literacy is associated with perceived benefits, which are in turn associated with ethical responsibility.

The high levels of ethical awareness observed ($M = 4.10$) are particularly noteworthy given concerns in the literature about AI misuse (Chan and Hu, 2023). Students in this study demonstrated strong commitment to responsible AI use ($M = 4.28$) and support for institutional guidelines ($M = 4.07$). This suggests that when students are given structured opportunities to engage with AI in meaningful ways, they develop not only skills but also a sense of ethical responsibility.

Practical implications

Based on the findings, here are some practical recommendations for teachers and schools:

- AI activities should be well-organized and guided, not left for students to figure out alone. In this study, students had to prepare lessons, present to classmates, and join discussions. This clear structure helped produce good results.
- Teachers should point out the real benefits of AI, such as saving time, improving work quality, and making learning more enjoyable. These benefits help students develop ethical awareness. Warning students only about the risks of AI may not work as well as showing them how AI can truly help them learn.

- Previous experience with AI matters. Students who had used AI before reported more benefits and higher ethical awareness. This means that schools should introduce AI activities early and give students repeated chances to use AI tools.
- Ethical rules should be made together with students. The finding that students strongly support clear guidelines ($M = 4.07$) shows they are ready to take part in creating AI policies. Schools should think about including students when making rules for AI use.

Limitations

This study has some limitations.

- First, the sample ($N = 104$) came from only one university, and students were chosen based on convenience. This means the results may not apply to all students.
- Second, the study used self-reported answers, which may not be completely accurate (common method bias; Podsakoff et al., 2003) because students might answer in a way they think looks good.
- Third, there was no control group, so we cannot say for sure that the activity caused the results.
- Fourth, the study measured outcomes right after the activity. We need follow-up studies to see if the gains in literacy, benefits, and awareness last over time.
- Fifth, although the study included open-ended questions, a systematic qualitative analysis was not conducted. Therefore, the study should be characterized as predominantly quantitative, with qualitative data used for exploratory purposes only. Future research should employ rigorous qualitative methods such as thematic analysis (Braun and Clarke, 2006) or in-depth interviews to capture richer insights into students' experiences.
- Sixth, the study's findings are based on self-reported data, which may be subject to social desirability bias or inaccurate recall. Future studies should incorporate objective measures of student performance and AI interaction behaviors (Podsakoff et al., 2003).

Recommendations for future research

Future studies should consider the following:

- Longitudinal studies to see if the effects continue over several semesters and lead to long-term changes in behavior.
- Control group studies comparing students who take part in structured AI activities with those who use AI without guidance or do not use AI at all.
- Direct measurement of co-creation behavior instead of relying only on what students report about themselves.
- Researchers could observe how students actually interact with AI tools or examine the work they produce.

- Studies across multiple institutions with larger and more diverse groups of students to make the findings more generalizable.

Examination of different conditions – for example, does the same mediation model work in different subjects (such as humanities versus science) or with different types of AI tools?

- Qualitative research exploring how students experience the shift from being consumers to co-creators. Although this study included some open-ended questions, in-depth interviews could provide deeper understanding of this change.

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Ethics statement

Ethical approval was not required for the studies involving humans because no prospective approval or waiver was obtained from a Research Ethics Committee prior to data collection as we don't have this committee in our university. However, the principals of code of ethics applied as the participation was voluntary, and students were informed that participation or non-participation would have no effect on their academic evaluation. To minimize potential instructor–student power dynamics, participation in the research component was separated from academic assessment, and students were informed of their right to decline participation without penalty. The collected data were handled confidentially and used solely for research purposes, with identifying information excluded from the analytical dataset. The author acknowledges the absence of prospective institutional ethical approval as a limitation and have disclosed this matter to the Editor. The studies were conducted in accordance with the local legislation and institutional requirements. Written informed consent for participation was not required from the participants or the participants' legal guardians/next of kin in accordance with the national legislation and institutional requirements because the survey was anonymous, participation was entirely voluntary, and no personally identifiable

information was collected. Participants were provided with information about the purpose and nature of the study before completing the questionnaire, and their decision to proceed with and submit the survey was considered implied informed consent.

Author contributions

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Generative AI statement

The author(s) declared that generative AI was not used in the creation of this manuscript.

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